StreetBite V1.0 Test Plan

StreetBite V1.0 is a bilingual Flutter app for discovering street food vendors, browsing meals from TheMealDB, saving favourites and placing orders, backed by Firebase. This plan describes how the app is tested manually, which screens are covered and how deeply, and when testing can be called finished. Login, Sign up and Home have fully designed test cases (41 in total, kept in StreetBite_Manual_QA.xlsx). The other screens get a shorter smoke checklist. Every failure is logged in Jira with severity, priority and a screenshot, and the Appium suite repeats the key flows on an emulator and a real phone.

1. Scope

In scope: splash and onboarding, Login, Sign up, Home with the live menu, Meal Details and favourites, Cart and orders, the Vendor Map, and the Profile screen with its fingerprint gate. English and Arabic, and light and dark themes, are checked wherever a screen is tested.

Out of scope: load and performance testing, security penetration testing, iOS, and the internals of Firebase and Google Maps themselves. Payment is simulated in the app, so only the order flow is tested, not a real gateway.

2. Objectives

- Confirm each requirement in the assignment brief behaves as written.
- Find defects before the beta testers do, and record each with evidence.
- Show which requirement every test case protects (see the Traceability sheet).
- Check that the app fails politely: no crashes on missing internet, bad input or interruptions.

3. Coverage by screen

Screen	What is verified	Test types	Depth
Splash	Fade-in animation, hand-off to Login or Home	GUI	Designed (TC_LOGIN_001)

Screen	What is verified	Test types	Depth
Login	Layout, valid and invalid credentials, empty fields, persistence, offline, double tap	GUI, positive, negative, general mobile	Designed (20 cases)
Sign up	Name, email, password and confirmation rules, duplicate email	Positive, negative	Designed (inside the Login suite)
Home	Banner, Hot Offers, two-column grid, add to cart, cache, theme and language	All four types	Designed (21 cases)
Meal Details	Content, add to cart, favourite toggle	Smoke	Checklist below
Cart	Totals, remove, empty state, payment choice, order placement	Smoke	Checklist below
Vendor Map	Six markers, info window with name, cuisine and rating	Smoke	Checklist below
Profile	Fingerprint prompt, photo, device details, favourites, order history, sign out	Smoke	Checklist below

Smoke checklists for the remaining screens

- **Meal Details:** opens with image, name, category, origin and instructions; Add to cart shows a SnackBar and raises the badge; the heart saves and removes a favourite; back returns to Home.
- **Cart:** items, quantities and total are correct; removing an item updates the total; emptying the cart shows a message; Cash and Card can be chosen; placing an order shows a success message and clears the cart.
- **Vendor Map:** the map loads with six markers in different cities; tapping a marker shows name, cuisine and rating; texts follow the selected language.
- **Profile:** the fingerprint prompt appears before the screen opens and a wrong or cancelled scan blocks it; photo picked from the gallery appears immediately; device model and Android version are shown; favourites and orders match what was saved;

sign out returns to Login and empties the cart.

4. Test types and approach

- **GUI:** layout, text, overflow, dark mode, Arabic and right-to-left.
- **Positive:** the normal path works with valid data.
- **Negative:** invalid input, wrong credentials, empty fields and unavailable services give a clear message and no crash.
- **General mobile:** no internet, backgrounding and resuming, rotation, double taps, system back button.
- **Automation:** an Appium and Java suite covers valid login, invalid credentials, empty fields, grid scrolling and add to cart with SnackBar. It is run on an Android emulator and on the physical phone, with screenshots saved for each test.

Cases are executed from the workbook, in suite order, on the physical phone first. Each case gets an actual result, a status (Pass, Fail or Blocked), the device and, for failures, a Jira key.

5. Environment

Item	Detail
Build	StreetBite 1.0.0+1, debug APK built with Flutter 3.35.6
Primary device	Samsung SM-A366B, real device, fingerprint enrolled
Secondary device	Android emulator
Backend	Firebase Authentication and Cloud Firestore (Spark plan), TheMealDB public API
Network	Wi-Fi for normal runs, airplane mode for offline cases
Tools	Jira for bugs, Appium 3 with UiAutomator2, Maven and JUnit 5, GitHub

6. Test data

A registered account, qa.tester@streetbite.test with password Test1234, is used for login, persistence and automation. Sign up cases use fresh addresses such as qa.new1@streetbite.test. Negative cases use nobody@streetbite.test, the password WrongPass123, the name ahmed and the password 12345.

7. Schedule and roles

This is a solo project, so one tester designs, executes and reports.

Date	Work
16 Oct	Plan, workbook and Appium suite ready, automation run on both devices
17 Oct	Execute all designed cases and the smoke checklists, log bugs in Jira
18 Oct	Retest fixes, complete the traceability matrix and the closure summary
19 Oct	Final check and submission

8. Entry and exit criteria

Entry: the app installs and launches on the physical phone, the test account exists in Firebase, internet is available and Firestore rules are published.

Exit (these are calculated on the Summary sheet):

- All designed cases have been executed.
- The pass rate of executed cases is at least 90%.
- No Critical or High defect is still open.
- Every failed case has a Jira key.
- Every requirement in the traceability matrix has at least one test case.

9. Defect management

Each defect is logged in Jira with steps to reproduce, expected and actual result, device, build, severity, priority and a screenshot. The same key is written in the Bug ID column of the failed case and in the DefectLog sheet.

Severity	Meaning
Critical	App crashes, data is lost, or login is impossible
High	A main feature is broken and there is no workaround
Medium	A feature works wrongly or only with a workaround
Low	Cosmetic issue or small text problem

Priority (Highest, High, Medium, Low) reflects how soon the fix is needed before submission, which can differ from severity. A fixed bug is retested on the same device before it is closed.

10. Deliverables

- This Test Plan.
- StreetBite_Manual_QA.xlsx with the test cases, the traceability matrix, the defect log and the closure summary.
- Jira bugs with severity, priority and screenshots.
- Appium test code and screenshots from the emulator and the physical phone.

11. Risks and assumptions

- **Error state is hard to reach.** Meals are cached after the first successful load, so the no-internet error screen only appears if the meal service is down on a first load. If it cannot be produced, that case is marked Blocked with the reason.
- **Biometric cases need an enrolled fingerprint.** Without one the app shows an unavailable message instead of the profile.
- **Emulator behaviour can differ from the phone.** The physical device result is the one that counts when they disagree.

- **One tester.** There is no independent review, so the traceability matrix is used to spot gaps.
- **Sample data.** Meal prices are derived from the meal id, and the six vendors on the map are hardcoded samples.